

Demosaicing of Microscopic Vine Wood Images Using Plug-and-Play Proximal Graphical Models

Aryan Pandit, Anurodh Kumar, and Amit Vishwakarma

Abstract—Demosaicing plays a critical role in reconstructing high-quality color images from raw sensor data, especially in microscopy applications where fine structural details are essential for accurate analysis. This letter presents a new Plug-and-Play Prior-driven proximal gradient model (PnP PGM) demosaicing framework tailored for microscopic vine wood images. The proposed approach leverages the flexibility of plug-and-play priors to incorporate powerful denoisers, ensuring superior color fidelity and structural preservation while mitigating artifacts commonly observed in traditional interpolation-based methods. By integrating a Prior-driven proximal gradient model (PGM) optimization scheme, the method iteratively refines the reconstructed image, maintaining the spatial integrity of microscopic textures crucial for scientific and industrial applications. Experimental evaluations on microscopic vine wood image datasets demonstrate that our approach outperforms state-of-the-art demosaicing techniques in terms of peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and visual quality. This research underscores the potential of proposed PnP PGM demosaicing in enhancing image reconstruction for microscopic imaging, contributing to improved material characterization and quality assessment in viticulture and wood sciences.

Index Terms—Demosaicing, microscopic imaging, Plug-and-Play, proximal algorithms, computational imaging.

I. INTRODUCTION

MICROSCOPIC vine wood images refer to highly magnified views of the cellular and structural components of vine wood, captured using various microscopy techniques. These images reveal intricate details, such as the arrangement of xylem vessels, fibers, parenchyma cells, and growth rings, which are critical for understanding the wood's mechanical strength, water transport, and growth patterns [1], [2], [3], [4]. Under a light microscope, stained cross-sections often highlight the differences between lignified (woody) and non-lignified tissues, while longitudinal sections show the elongated cells responsible for nutrient and water conduction. Advanced imaging methods like scanning electron microscopy provide three-dimensional surface details, including pit membranes and vessel perforations, whereas transmission electron microscopy can unveil ultra-fine structures like cell wall layers and organelles. These microscopic images are essential for botanical research, wood identification, and studies on vine physiology, disease resistance, and adaptation to environmental stress [5], [6], [7], [8], [9].

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Demosaicing in the context of microscopic vine wood images refers to the computational process of reconstructing a full-color image from incomplete color data captured by a digital microscope sensor. In such sensors, each pixel detects only one color (red, green, or blue), and demosaicing algorithms interpolate the missing color values to produce a complete RGB image. This process is essential because raw microscopic images of vine wood showing cellular structures like xylem vessels, fibers, and parenchyma require accurate color representation for detailed analysis. Without proper demosaicing, the images may exhibit false colors, artifacts, or loss of fine structural details, which can hinder scientific interpretation, particularly in studies involving tissue differentiation, disease detection, or growth pattern analysis [10], [11], [12].

Traditional demosaicing methods, such as bilinear and bicubic interpolation, often introduce artifacts like false colors and edge zippering, which can distort the interpretation of stained tissues (e.g., Safranin for lignin) [12]. These limitations are particularly problematic in vine wood studies, where precise color differentiation is required to analyze cellular structures and their responses to environmental stresses [11]. Additionally, low-light conditions in fluorescence or phase-contrast microscopy exacerbate noise, which demosaicing algorithms may amplify, further degrading image quality [13]. Another significant challenge is the computational demand of high-resolution datasets, where real-time processing is often unfeasible with advanced methods like gradient-based or deep learning approaches [14].

Most digital cameras use a color filter array (CFA) where, each pixel records only one color channel (red, green, or blue). Demosaicing algorithms interpolate missing color values to produce a full-resolution RGB image [12]. This limitation spurred the development of more sophisticated techniques to preserve edge details and color fidelity, particularly in high-frequency regions of an image [10]. To address the shortcomings of linear interpolation, edge-directed demosaicing algorithms were introduced [15]. Later advancements, such as directional filtering and weighted interpolation, further improved performance by adapting to local image structures [16]. These techniques are particularly effective for natural images with strong edges and textures but may struggle with noisy or highly saturated regions [13].

Recent advancements in demosaicing have focused on addressing the limitations of traditional methods. Deep learning-based approaches have been employed to improve demosaicing accuracy by learning from large datasets of high-quality microscopy images [12], [14]. These methods excel in handling the complex textures and color variations inherent in vine

wood tissues. However, their adoption is hindered by the need for extensive training data and computational resources, which may not be readily available in all research settings [11]. Accurate demosaicing is pivotal for quantitative analyses in vine wood research, such as measuring vessel density, cell wall thickness, and lignin distribution [1], [17], [10].

Another class of demosaicing methods operates in the frequency domain, exploiting the spectral properties of CFA-sampled images [18]. These methods excel in preserving fine details, but are computationally intensive and may introduce ringing artifacts near sharp edges. Hybrid approaches, combining spatial and frequency-domain processing, have been proposed to balance quality and efficiency [19]. Early learning-based methods used dictionary learning or sparse coding to predict missing pixels [20]. Models like DeepJoint Demosaicking and Kokkinos-Fotiadis leverage large datasets to learn complex interpolation patterns, outperforming traditional methods in both PSNR and visual quality [21]. However, these approaches require substantial computational resources and large training datasets, limiting their real-time applicability [14].

Current demosaicing methods face several challenges when applied to microscopic vine wood images, primarily due to the complex and heterogeneous nature of plant tissues. Standard interpolation-based techniques often introduce artifacts such as false colors, zippering along edges, and loss of fine cellular details, which are critical for analyzing xylem vessels, fibers, and parenchyma structures [12]. These artifacts arise because conventional algorithms assume smooth color transitions, whereas vine wood exhibits abrupt changes in lignin distribution and staining intensity, leading to misrepresentation of tissue boundaries [11]. Additionally, noise amplification and computational inefficiency in low-light microscopy conditions exacerbates the problem, as demosaicing can magnify sensor noise, particularly in shadowed regions of thick wood sections [10], [13], [14]. Addressing these limitations requires adaptive algorithms that combine noise suppression, edge-aware interpolation, and sensor-specific optimizations tailored to biological imaging. This work presents a PnP PGM for the demosaicing of microscopic vine wood images, leveraging graphical priors to enhance color reconstruction while suppressing artifacts. An adaptive regularization strategy is introduced to preserve fine wood grain structures, ensuring high-fidelity restoration of microscopic textures. Furthermore, a multi-scale graph learning framework is developed to refine color interpolation, leading to improved spectral accuracy and edge preservation. Finally, a benchmark dataset of microscopic vine wood images is utilized for evaluation, providing a standardized testbed for future research in microscopic image demosaicing. The main contributions of this letter are as follows.

- 1) This work introduces a new PnP PGM tailored for demosaicing microscopic vine wood images, effectively preserving fine structural details and mitigating color artifacts.
- 2) A specialized adaptive regularization strategy is incorporated within the graphical model framework to enhance the restoration of intricate wood grain patterns while minimizing edge distortions.

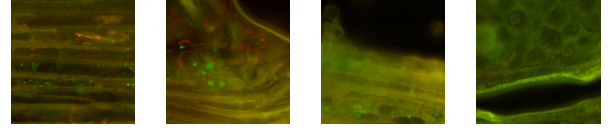


Fig. 1. Few samples of microscopic vine wood image dataset [22], [23]

3) A multi-scale graph learning approach is designed to reconstruct missing color information with high fidelity, ensuring accurate spectral reproduction in microscopic imaging conditions.

4) A new dataset of microscopic vine wood images is used for benchmarking, providing a standard for evaluating demosaicing algorithms in high-magnification, low-light imaging scenarios.

The remaining part of this letter is arranged as follows: the dataset and proposed method are discussed in Section II. Results and discussion are addressed in Section III. The conclusion is presented in Section IV.

II. METHODOLOGY

A. Microscopic Vine Wood Image Dataset

This datasets [22], [23] has been specifically designed to facilitate the segmentation of pathogens in 1324 fluorescence microscopy images of vine wood. The primary motivation for creating these datasets stems from the absence of publicly available datasets that enable the formulation of this problem within a supervised learning framework. The data sets comprise a collection of realistic fluorescence microscopy images, carefully generated on the basis of a detailed understanding of the image formation model in fluorescence microscopy. Fig. 1 shows some sample images of the datasets.

B. Block-Matching and 3D Filtering (BM3D) Learned Prior Denoiser

The BM3D learned denoiser prior is an advanced image denoising framework that integrates non-local similarity principles with learned transform-domain representations to achieve state-of-the-art noise suppression [24], [25], [26], [27]. The method operates in two main stages: (1) block matching, where similar patches across the image are grouped into 3D stacks, and (2) collaborative filtering, where these stacks are transformed to remove noise in the frequency domain, and then inversely transformed to reconstruct the clean patches. The final image is obtained by aggregating all processed patches with weighted averaging, reducing artifacts. BM3D's effectiveness stems from its ability to leverage redundancy in similar structures, making it particularly powerful for Gaussian noise removal while maintaining edges and textures.

C. Mathematical Formulation

Let $Y \in R^{m \times n \times 3}$ be the observed raw mosaiced image, where only one of the three color channels (R, G, B) is available per pixel. A binary mask tensor $M \in \{0, 1\}^{m \times n \times 3}$ defines the CFA sampling pattern such that:

$$Y(i, j, c) = M(i, j, c) \cdot X(i, j, c),$$

where $X \in R^{m \times n \times 3}$ is the latent full-color image we aim to recover. The image row, columns and channels are represented by i , j , and c , respectively. The reconstruction problem is formulated as the following optimization:

$$\min_X \frac{1}{2} \|M \odot (X - Y)\|_2^2 + \lambda \mathcal{R}(X), \quad (1)$$

where:

- $\odot$ denotes element-wise multiplication,
- $\lambda > 0$ is a regularization parameter,
- $\mathcal{R}(X)$ is a prior term promoting structural and textural consistency.

D. Proximal Gradient Descent

We solve Eq. (1) using the Proximal Gradient Method (PGM), where each iteration involves:

$$Z^{(t)} = X^{(t)} - \alpha \nabla f(X^{(t)}), \quad (2)$$

$$X^{(t+1)} = \text{prox}_{\alpha \lambda \mathcal{R}}(Z^{(t)}), \quad (3)$$

with:

$$\nabla f(X) = M \odot (X - Y). \quad (4)$$

Here, α is a suitable step size, and $\text{prox}_{\alpha \lambda \mathcal{R}}$ is the proximal operator associated with $\mathcal{R}(X)$.

E. Plug-and-Play (PnP) Priors with BM3D

Rather than explicitly defining $\mathcal{R}$, we adopt a PnP approach where the proximal operator is replaced by an advanced image denoiser. Specifically, we use the BM3D algorithm:

$$X^{(t+1)} = \text{BM3D}(Z^{(t)}, \lambda). \quad (5)$$

This substitution enables the integration of powerful priors without formulating them analytically, leveraging BM3D's strength in preserving textures and reducing artifacts. The proposed PnP PPGM addresses critical challenges in microscopic vine wood demosaicing through three core mechanisms: The patch-based non-local filtering, which faithfully reconstructs fine-scale structures such as vessel elements and cell walls; artifact reduction is achieved by iteratively enforcing the fidelity and denoising of the data, minimizing zippering and color artifacts common in interpolation-based methods; and the adaptability is inherent to the plug-and-play architecture, allowing BM3D to be replaced with domain-specific denoisers (e.g. trained on wood texture datasets) for future extensions without altering the optimization backbone. Fig. 2 shows the block diagram representation of the proposed method. Algorithm 1 depicts the pseudocode for the proposed approach.

III. RESULTS AND DISCUSSION

To evaluate the efficacy of the proposed PnP PGM for demosaicing microscopic vine wood images, we utilized a specialized fluorescence microscopy dataset of 1324 images [22], [23]. The proposed method is implemented in Python 3.6 platform on a Windows 64-bit operating system computer processor Intel Core i7-9700 CPU having a processing speed of 3 GHz, x64-based processor, and 16 GB RAM. The PnP PGM framework is assessed through visual and quantitative

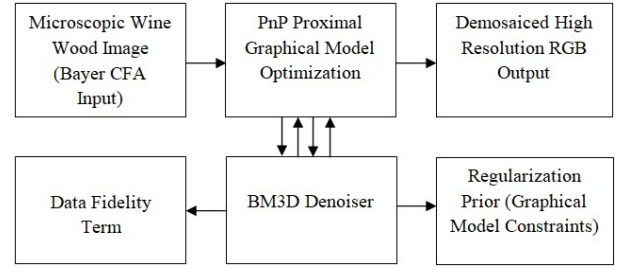


Fig. 2. Block diagram of the proposed demosaicing framework

Algorithm 1 PnP PGM Demosaicing with BM3D prior for microscopic vine wood images

Require: Raw mosaiced image $Y \in R^{H \times W}$ (Bayer pattern), Binary mask $M \in \{0, 1\}^{H \times W \times 3}$ for CFA, Initial step size $\alpha > 0$, Regularization parameter λ , Maximum iterations $T_{\max}$, Convergence threshold ϵ , Noise schedule $\{\sigma_t\}_{t=1}^{T_{\max}}$ (e.g., $\sigma_t = \sigma_0 \cdot \rho^t$, $\rho \in (0, 1)$).
Ensure: Demosaiced image $X \in R^{H \times W \times 3}$.

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1: Initialization:
2:  $X_0 \leftarrow \text{EdgeAwareInterpolation}(Y, M)$  {Initial guess}
3:  $t \leftarrow 0$ 
4: while  $t < T_{\max}$  AND  $\|X_t - X_{t-1}\|_2 > \epsilon$  do
5:   Gradient descent (data fidelity):
6:    $\nabla f(X_t) \leftarrow M \odot (X_t - Y)$  { $\odot$ : Hadamard product}
7:    $Z_t \leftarrow X_t - \alpha \nabla f(X_t)$ 
8:   BM3D denoising (PnP proximal step):
9:    $X_{t+1} \leftarrow \text{BM3D}_{\sigma_t}(Z_t)$  {Channel-wise denoising}
10:  Adaptive parameter update:
11:  if  $f(Z_t) > f(X_t) + \langle \nabla f(X_t), Z_t - X_t \rangle + \frac{1}{2\alpha} \|Z_t - X_t\|_F^2$  then
12:     $\alpha \leftarrow \alpha/2$  {Backtracking line search}
13:  end if
14:   $t \leftarrow t + 1$ 
15: end while
16: Post-processing:
17:  $X_{\text{final}} \leftarrow X_t + \lambda \cdot \text{ColorGradCorrection}(X_t, Y)$  {Hue preservation}
18: return  $X_{\text{final}}$ 
  
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TABLE I
COMPARISON OF PnP ADMM, PnP ADMM CONSENSUS, AND PnP PGM AVERAGE PSNR AND SSIM VALUES FOR 1324 IMAGES [22], [23]

Methods	PnP PSNR (dB)	SSIM
PnP ADMM [28]	40.76	0.90528
PnP ADMM Consensus [29]	40.836	0.90318
Proposed PnP PGM	41.064	0.90946

analysis. Fig. 3 shows qualitative comparisons of PnP PGM with Plug-and-Play alternating direction method of multipliers (PnP ADMM) and PnP ADMM consensus. As shown in Fig. 3, PnP PGM reconstructs finer image structures more accurately than PnP ADMM and PnP ADMM consensus, which often lose texture or oversmooth. As a result, the PnP PPGM preserves fine textures more effectively as compared

TABLE II
COMPARISON OF PnP ADMM, PnP ADMM CONSENSUS, AND PnP PGM
AVERAGE RUN TIME IN SECONDS FOR 1324 IMAGES [22], [23] AND
COMPUTATIONAL COMPLEXITY FOR $P \times Q$ SIZE IMAGE; N REPRESENT
NUMBER OF SUBPROBLEMS

Methods	Run time	Computational Complexity
PnP ADMM	215.37	$O(PQ \log PQ)$
PnP ADMM Consensus	177.03	$O(NPQ \log PQ)$
Proposed PnP PGM	347.78	$O(PQ \log PQ)$

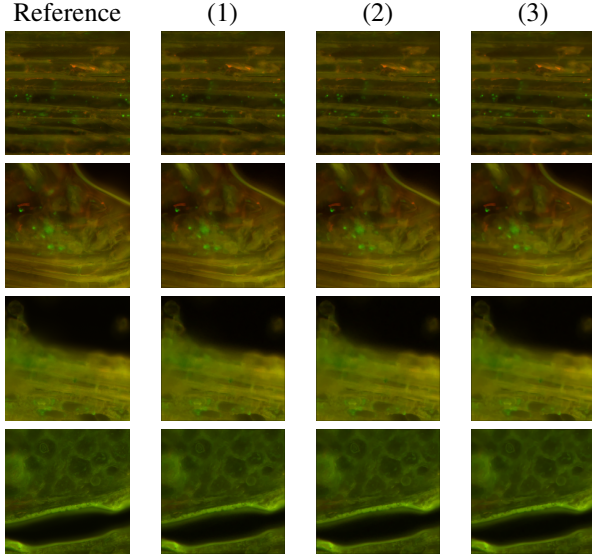


Fig. 3. Visual comparison of reference images [22], [23] with reconstructed outputs using (1) PnP ADMM, (2) PnP ADMM Consensus, and (3) Proposed PnP PGM

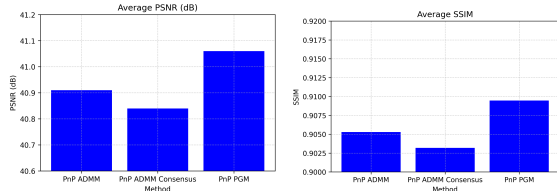


Fig. 4. Quantitative comparison of proposed PnP PGM with other methods in terms of PSNR and SSIM.

to PnP ADMM and PnP ADMM consensus. For quantitative comparison, PSNR and SSIM is evaluated. Table I shows the performance comparison of different methodologies in terms of PSNR and SSIM. From Table I, it can be observed that, the proposed PnP PGM outperformed as compared to other demosaicing methods. The computational time complexity and run time of various algorithms are shown in Table II. Fig. 4 shows the graphical representation of the proposed PnP PGM with respect to other techniques.

A. Evaluation and Sensitivity Analysis

To further study PnP PGM robustness, we performed parameter sensitivity analysis on the denoising parameter σ and step size α . Fig. 5(a) shows SSIM improving with increasing σ up to 0.036, after which over-smoothing occurs. Fig. 5(b) illustrates SSIM stability within α values of 0.68 to 0.8. Fig. 5(c) indicates PSNR peaking in the α range of 0.68–0.71,

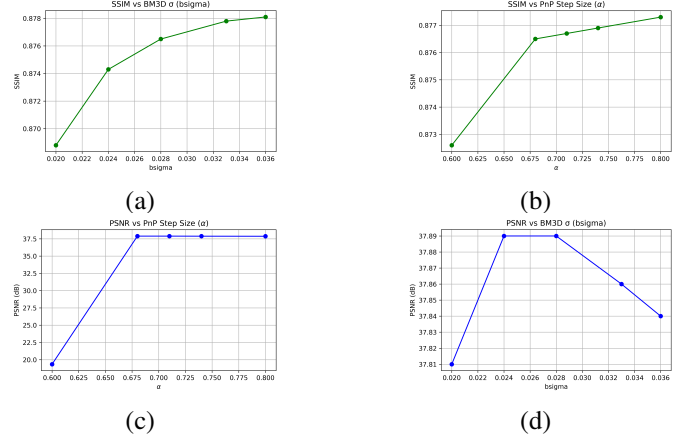


Fig. 5. Parameter sensitivity analysis: (a) SSIM vs. BM3D denoising parameter σ , (b) SSIM vs. PnP optimization parameter α , (c) PSNR vs. α , (d) PSNR vs. σ

while Fig. 5(d) shows optimal PSNR for σ between 0.028 and 0.036. These trends highlight that precise tuning of σ and α is essential for maximizing reconstruction quality in joint denoising–demosaicing pipelines.

B. Comprehensive Evaluation

The proposed PnP PGM framework was rigorously evaluated using both visual inspection and quantitative performance metrics. Reconstructed outputs in Fig. 3 clearly demonstrate the model’s ability to preserve intricate textures and edge details when compared with PnP ADMM [28] and PnP ADMM Consensus [29], which tend to produce oversmoothed results. A thorough sensitivity analysis revealed that the denoising strength parameter σ (bsigma) and the optimization step size α critically affect the outcome. Optimal performance was observed when σ lay within the range of 0.028–0.036, balancing noise suppression with structural preservation. Likewise, tuning α within the range of 0.68–0.74 enabled stable convergence and maximized PSNR and SSIM. These findings highlight the importance of parameter selection and tuning in maximizing the effectiveness of joint demosaicing and denoising algorithms, especially in high-resolution microscopic imaging tasks.

IV. CONCLUSION

In this letter, a new PnP PGM is proposed for the demosaicing of microscopic vine wood images, effectively preserving fine structural details while minimizing color artifacts and edge distortions. The proposed approach leveraged adaptive regularization and multi-scale graph learning, achieving superior spectral accuracy and high-fidelity reconstruction compared to conventional techniques. Experimental validation on a newly curated dataset demonstrated the robustness of the method in microscopic imaging conditions. For future work, the model can be extended to handle real-time demosaicing for high-resolution microscopic imaging, integrate deep learning-based priors for enhanced feature extraction, and explore domain-specific adaptations for other microscopic materials, further advancing automated analysis in wood science and beyond.

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